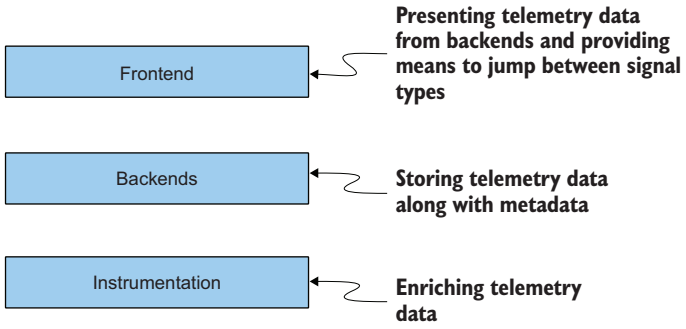


Figure 11.1 The correlation stack

The correlation stack, from bottom to top, consists of the following:

- *The instrumentation layer*—This layer comprises your code and the telemetry agent, enriching the telemetry data and ingesting it into backends. We covered this in detail in chapters 3 and 4. Note that without the support of this layer, the higher layers are challenging to implement. Many of the sources-related metadata pieces (e.g., which lambda function or pod in which namespace in which Kubernetes cluster emitted a, say, span) are only available at collection time, and it is hard to provide comprehensive correlation without said information.
- *The backends layer*—Telemetry data is stored along with metadata in this layer. This may consist of separate backends per signal type or backends with several signal types, such as OpenSearch. See also chapter 5, where we covered this topic in greater detail. This layer is mostly concerned with driving the next layer, offering the raw data that is then presented to the user for choosing a concrete correlation path.
- *The frontend layer*—Sometimes also referred to as the *presentation layer* (see chapter 6, where we covered the topic, including tooling), this layer is where signals are presented and can be used to consume correlated telemetry data. Various visualizations and renderings can help you quickly select, filter, and jump between signal types.

The metadata necessary to drive correlation can either be standardized after the fact (e.g., when using the Prometheus relabeling feature discussed in chapter 4) or up front. The latter approach, establishing defined semantics, is what OpenTelemetry does with its semantic conventions. Let's have a closer look at that now.

11.1.1 Correlation with OpenTelemetry

With semantic conventions (https://opentelemetry.io/docs/concepts/semantic-conventions/), OpenTelemetry defines a metadata scheme, including common names and sometimes common values for different kinds of telemetry data. More specifically, OpenTelemetry supports the following metadata to enable signal correlation in backends: